

B.Sc. (Data Science)

DS101T : Problem Solving and Python Programming

Unit 4: GUI Programming

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Unit4: GUI Programming

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GUI

- GUI helps user to interact with computers.
- Python supports various libraries to support GUI:
 - Kivy
 - Python Qt
 - wxPython
 - Tkinter

Python Tkinter

- Python Tkinter is a built-in Python library for creating graphical user interfaces (GUIs).
- It includes various tools and widgets for developing desktop applications with graphical interfaces.
- Since Tkinter comes with most Python installations, developers can easily use it without needing additional libraries or installations.
- The name "Tkinter" stands for "Tk interface," indicating its basis on the Tk GUI toolkit.
- Tkinter enables the creation of windows, buttons, labels, text boxes, and other GUI components to develop interactive applications.
- It provides a fast and easy way to create desktop applications.

- Tkinter provides a variety of widgets like buttons, labels, text boxes, menus and more that can be used to create interactive user interfaces.
- Tkinter supports event-driven programming, where actions are taken in response to user events like clicks or keypresses.

Uses of Tkinter

1. **Creating windows and dialog boxes:** Tkinter can be used to create windows and dialog boxes that allow users to interact with your program. These can be used to display information, gather input, or present options to the user.
2. **Building a GUI for a desktop application:** Tkinter can be used to create the interface for a desktop application, including buttons, menus, and other interactive elements.
3. **Adding a GUI to a command-line program:** Tkinter can be used to add a GUI to a command-line program, making it easier for users to interact with the program and input arguments.
4. **Creating custom widgets:** Tkinter includes a variety of built-in widgets, such as buttons, labels, and text boxes, but it also allows you to create your own custom widgets.
5. **Prototyping a GUI:** Tkinter can be used to quickly prototype a GUI, allowing you to test and iterate on different design ideas before committing to a final implementation.

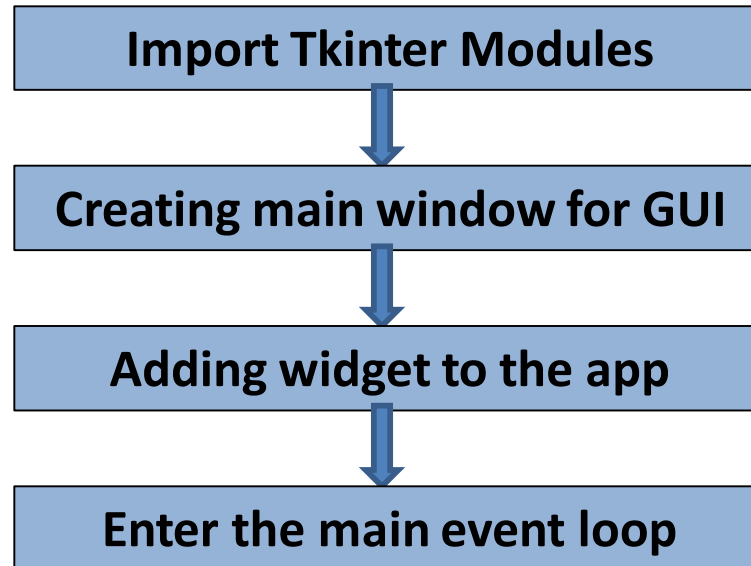
Installing Tkinter

- Tkinter can be installed using pip. The following command is run in the command prompt to install Tkinter.

`pip install tk`

- This command will start downloading and installing packages related to the Tkinter library. Once done, the message of successful installation will be displayed.

Fundamental structure of Tkinter program



Tkinter Programming

- Setup a Tkinter Application:
 - Import the Tkinter module.
 - Create a main application window.
 - Run the Tkinter event loop.

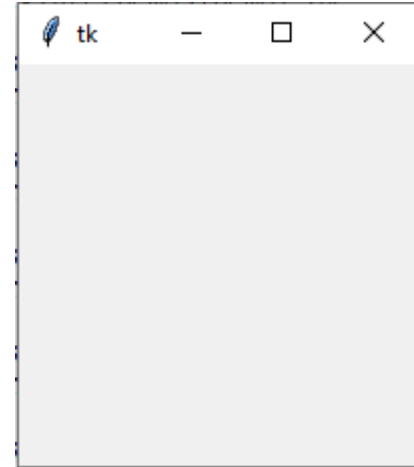
First Tkinter GUI

```
import tkinter as tk

window = tk.Tk()

#Widgets are added here

window.mainloop()
```



Where,

`window = tk.Tk()` will create the main application window

`window.mainloop()` to Run the Tkinter event loop continuously till user cannot close the window explicitly.

```
import tkinter as tk
```

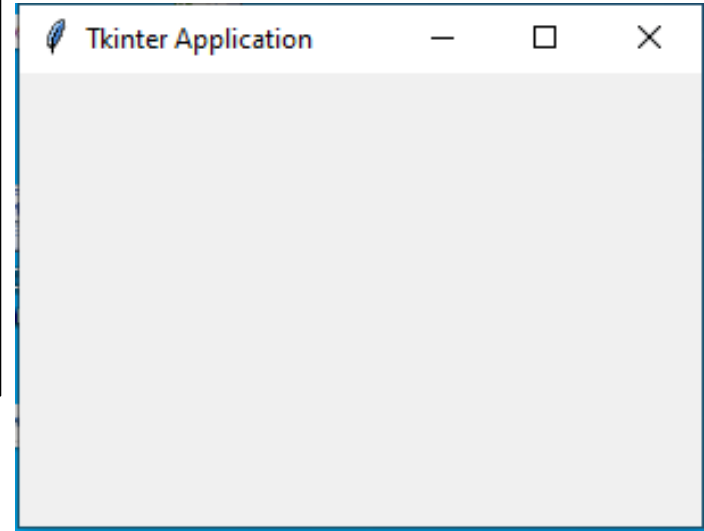
```
# Create the main application window
```

```
window = tk.Tk()
```

```
# Run the Tkinter event loop
```

```
window.mainloop()
```

Output:



```
from tkinter import *
```

```
window=Tk()
```

```
window.title("My First Tkinter Application")
```

```
window.mainloop()
```

Tkinter window

➤ **Changing the window title:**

```
window.title("****Add Title here****")
```

➤ **Changing the default icon of window:**

```
window.iconbitmap(r"Add Absolute path of icon")
```

r defines absolute path

➤ **Setting widow background color:**

```
window['bg']="Colorname"
```

OR

```
window.config(bg="Colorname")
```

➤ **Making the window transparent:**

```
window.attributes('-alpha', 0.5)
```

Transparency Range vary between 0 and 1:

0 means more transperent and 1 means more visible.

Tkinter window Demonstration

```
from tkinter import *  
window=Tk()
```

```
window.title("My First Tkinter application")
```

```
window.iconbitmap(r"F:\reshma\TkinterDemos\PythonLogo.png")
```

```
window.attributes('-alpha',0.5)
```

```
window['bg']="Red"
```

```
window.mainloop()
```

Tkinter Frame

- The Frame widget is very important for the process of grouping and organizing other widgets in a somehow friendly way. It works like a container, which is responsible for arranging the position of other widgets.
- It uses rectangular areas in the screen to organize the layout and to provide padding of these widgets. A frame can also be used as a foundation class to implement complex widgets.
- **Syntax: `w = frame( master, options)`**
- **Parameters:**
 - master: This parameter is used to represent the parent window.
 - options: There are many options which are available and they can be used as key-value pairs separated by commas.

```
#Tkinter aplicacion to create frame
```

```
from tkinter import *
```

```
window=Tk()
```

```
window.title("Tkinter Label")
```

```
window.geometry("500x500")
```

```
frame = Frame(window, bg="blue",borderwidth=6)
```

```
frame.pack(side=LEFT)
```

```
l1=Label(frame, text="My First Tkinter Label",bg="Yellow",fg="Red")
```

```
l1.pack()
```

```
window.mainloop()
```

Tkinter widgets

- In Tkinter, a widget is essentially a graphical component that the user can interact with.
- They can range from simple elements like buttons and labels to more complex ones like text entry fields, listboxes, and canvases.
- Each widget serves a specific purpose and can be customized to fit the design and functionality requirements of your application.
- Each widget in Tkinter is an instance of a specific class defined in the Tkinter Module.
- These classes provide methods and attributes that allow you to configure the widget's appearance, behavior, and functionality.
- Widgets are typically added to the application's window or frames using layout managers like `pack()`, `grid()`, or `place()`, which determine their position and size within the interface.

Tkinter Widget Basics

Widget	Description
Label	Display static text or images.
Button	It is used to add buttons to your application
Entry	It is used to input single line text entry from user.
Frame	It is used as container to hold and organize the widgets.
RadioButton	It is used to implement one-of-many selection as it allows only one option to be selected.
Checkbutton	Create checkboxes for boolean options.
ListBox	Display a list of items for selection.
Scrollbar	Add scrollbars to widgets like Listbox.
Menu	It is used to create all kinds of menu used by an application.
Canvas	Draw shapes, lines, text, and images.
Text	Create a multiline text input with advanced editing capabilities.
ComboBox	Provide a dropdown list with editable text entry.
Message	Display simple messages or notifications.
MessageBox	Display dialog boxes for messages, warnings, etc.